

S. A. I. Shouborno

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Research interests

Interpretable and physics-informed machine learning for sensor data; wearable health sensing, including cuffless blood pressure and sleep-based cognitive screening; industrial cyber-physical systems; language models grounded in sensor data; efficient on-device inference.

Education

Ph.D., Electrical and Computer Engineering, University of Massachusetts Amherst Aug 2023 – May 2028 (expected)
Advisor Bashima Islam. Began at Worcester Polytechnic Institute in Aug 2023; transferred with the lab in 2026.

M.S., Electrical and Computer Engineering, Worcester Polytechnic Institute Aug 2023 – Dec 2025
GPA 4.00/4.00. Thesis: Interpretable and Efficient AI for Sensor-Rich Mobile and Embedded Systems.

B.Sc., Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology Feb 2016 – Feb 2021
Thesis: Audiovisual emotion recognition using multi-head attention based neural network, advised by Celia Shahnaz.

Experience

Research Assistant, University of Massachusetts Amherst, Amherst, MA Sep 2026 – Present

- Building physics-informed neural networks that estimate blood pressure from graphene electronic tattoo bioimpedance, with the Kireev Lab.
- Extending the Bosch internship project with Bosch Research by reconstructing the program execution traces of industrial controllers from process sensors (IAAI-27, under review). The method detects code-path faults that sensor readings alone cannot reveal at 0.99 AUC and names the faulty function 69% of the time.
- Predicted future cognitive impairment from overnight sleep studies at 0.77 age-conditioned AUROC on held-out hospitals in the PhysioNet Challenge 2026 (CinC 2026, accepted).

AI for Cyber-Physical Systems Intern, Robert Bosch LLC, Bosch Research, Pittsburgh, PA May 2025 – Aug 2025

- Built a contrastive model aligning vision, sensor, program trace and text data from a manufacturing process for language-guided few-shot classification of machine state.

Research and Teaching Assistant, Worcester Polytechnic Institute, Worcester, MA Aug 2023 – Aug 2026

- Developed LLaSA, a sensor-aware LLM that answers questions about human activity from IMU data and outperforms GPT-3.5 and GPT-4o-mini, with 235,661 released training pairs (SensorCap and OpenSQA).
- Developed JOLT, an early-exit training method that ranked first of 11 methods across 7 datasets under compute budgets.
- Served as teaching assistant for the graduate course On-Device Deep Learning and three undergraduate ECE courses.

Machine Learning Engineer, Hishab Technologies, Dhaka, Bangladesh Feb 2021 – Jul 2023

- Built telephony conversational AI for Bengali speakers, covering speech recognition, text to speech, named entity recognition, language models and grapheme-to-phoneme conversion.
- Deployed services for banking and microfinance clients, including Bank Asia and Hitachi India.

Publications

Under review

S. A. I. Shouborno, Emily Ruppel, Sudershan Boovaraghavan, Bashima Islam. **Program Execution Traces Without Instrumentation: Debugging Industrial Controllers from Physical Process Sensors**. AAAI Conference on Innovative Applications of Artificial Intelligence (IAAI-27). [code]

Journal articles

T. I. Mahmud, S. A. Imran, C. Shahnaz. **Res-SE-ConvNet: A Deep Neural Network for Hypoxemia Severity Prediction for Hospital In-Patients Using Photoplethysmograph Signal**. IEEE Journal of Translational Engineering in Health and Medicine, vol. 10, pp. 1–9, 2022. [code]

A. B. A. Qayyum, K. M. N. Hassan, A. Anika, M. F. Shadiq, M. M. Rahman, M. T. Islam, S. A. Imran, S. Hossain, M. A. Haque. **DOANet: A Deep Dilated Convolutional Neural Network Approach for Search and Rescue with Drone-Embedded Sound Source Localization**. EURASIP Journal on Audio, Speech, and Music Processing, vol. 2020, no. 1, art. 16, 2020. [code]

Conference papers

- S. A. I. Shouborno, Mohammad Nur Hossain Khan, Bashima Islam. **JOLT: Jointly Optimized Layerwise Termination for Budgeted Early-Exit Inference**. IEEE International Conference on Tools with Artificial Intelligence (ICTAI). **Accepted, 2026**. [\[code\]](#)
- S. A. I. Shouborno, Bashima Islam. **Cross-Site Generalization of Interpretable Sleep Features for Predicting Future Cognitive Impairment**. Computing in Cardiology (CinC). **Accepted, 2026**. [\[code\]](#)
- S. A. Imran, M. T. Islam, C. Shahnaz, M. T. Islam, O. T. Imam, M. Haque. **COVID-19 mRNA Vaccine Degradation Prediction Using Regularized LSTM Model**. IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE), pp. 328–331, 2020. [\[code\]](#)
- O. T. Imam, M. Haque, C. Shahnaz, S. A. Imran, M. T. Islam, M. T. Islam. **Detection of Tuberculosis from Chest X-Ray Images Based on Modified Inception Deep Neural Network Model**. IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE), pp. 360–363, 2020.
- M. T. Islam, S. A. Imran, A. Arefeen, M. Hasan, C. Shahnaz. **Source and Camera Independent Ophthalmic Disease Recognition from Fundus Image Using Neural Network**. IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON), pp. 59–63, 2019.
- A. B. A. A. Qayyum, A. Anika, M. M. M. Miah, M. M. Rahman, K. M. N. Hasan, M. T. Islam, S. A. I. Shouborno, M. F. Shadiq, M. A. Haque. **Direction of Arrival Estimation through Noise Suppression: A Novel Approach Using GSC Beamforming and Room Acoustic Simulation**. IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON), pp. 104–108, 2019.
- S. A. I. Shouborno, T. I. Mahmud, N. Ishraq, R. Ali, T. H. Joy, S. A. Fattah, A. Bhattacharjee, C. Shahnaz. **Complete Automation of an E-commerce System with Internet of Things**. IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON), pp. 81–86, 2019.

Workshop and short papers

- Tyler Bliss, Mahit Verma, Aila Iyer-Singh, Subrata Biswas, **Sheikh Asif Imran**, Bashima Islam. **Concord: An Agreement-Aware Multi-Adjudication Pipeline for LLM Evaluation**. Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM), pp. 502–510, 2026.
- S. A. I. Shouborno, Mohammad Nur Hossain Khan, Subrata Biswas, Bashima Islam. **LLaSA: A Sensor-Aware LLM for Natural Language Reasoning of Human Activity from IMU Data**. Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion), pp. 893–899, 2025. [\[code and datasets\]](#)

Patent

T. Al Muntasir, M. T. Islam, S. A. Imran. **System and Method for Optimizing a User Interaction Session within an Interactive Voice Response System**. U.S. Patent Application No. 18/378,118, 2025.

Honors and awards

2nd place (ICIP 2019) and 5th place (ICIP 2020), IEEE Video and Image Processing Cup	2019, 2020
4th place, IEEE Signal Processing Cup at ICASSP 2020	2020
National champion and 22nd worldwide, IEEEExtreme 12.0 programming contest (also national champion in 2017)	2018

Service

Chair, IEEE Signal Processing Society Student Chapter, BUET	2019 – 2021
- Organized the 2019 IEEE SPS Winter School on Multimodal Signal Processing. - Grew it into the largest IEEE student branch chapter in Bangladesh, earning an IEEE SPS growth reward for membership growth.	
Ambassador and Country Lead (2019), IEEEExtreme Programming Contest	2017 – 2019

Skills

Programming: Python, PyTorch, TensorFlow, Keras, scikit-learn, MATLAB, C/C++, Git, LaTeX

Machine learning: Multimodal and contrastive learning, LLM fine-tuning, on-device inference, physics-informed models

Signals: Biomedical signals (PPG, bioimpedance, sleep PSG), IMU, audio and speech, image and video

Systems: SLURM clusters, MLOps, GCP and AWS, Kafka, Jenkins, microcontrollers, FPGA

References

Available on request.